

5 Telomeres are lengths of DNA that consist of repetitive nucleotide sequences. Telomeres are present in eukaryotic chromosomes.

(a) Outline the role of telomeres in eukaryotic chromosomes.

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..... [3]

(b) Fig. 5.1 shows part of the telomere nucleotide sequence in one of the DNA strands.

A A T C C C A A T C C C A A T C C C

Fig. 5.1

Scientists have found that the DNA in telomeres can be transcribed to produce RNA known as TERRA.

(i) TERRA is transcribed from the DNA nucleotide sequence shown in Fig. 5.1.

Complete Fig. 5.2 to show the six bases in the RNA sequence of TERRA.

DNA sequence:	A	A	T	C	C	C
RNA sequence:						

Fig. 5.2

[1]

- (ii) Fig. 5.3 is a diagram showing DNA triplet codes on the non-transcribed strand of a gene and the amino acids coded by the triplets.

The bases in the centre of the diagram represent the first base in a triplet.

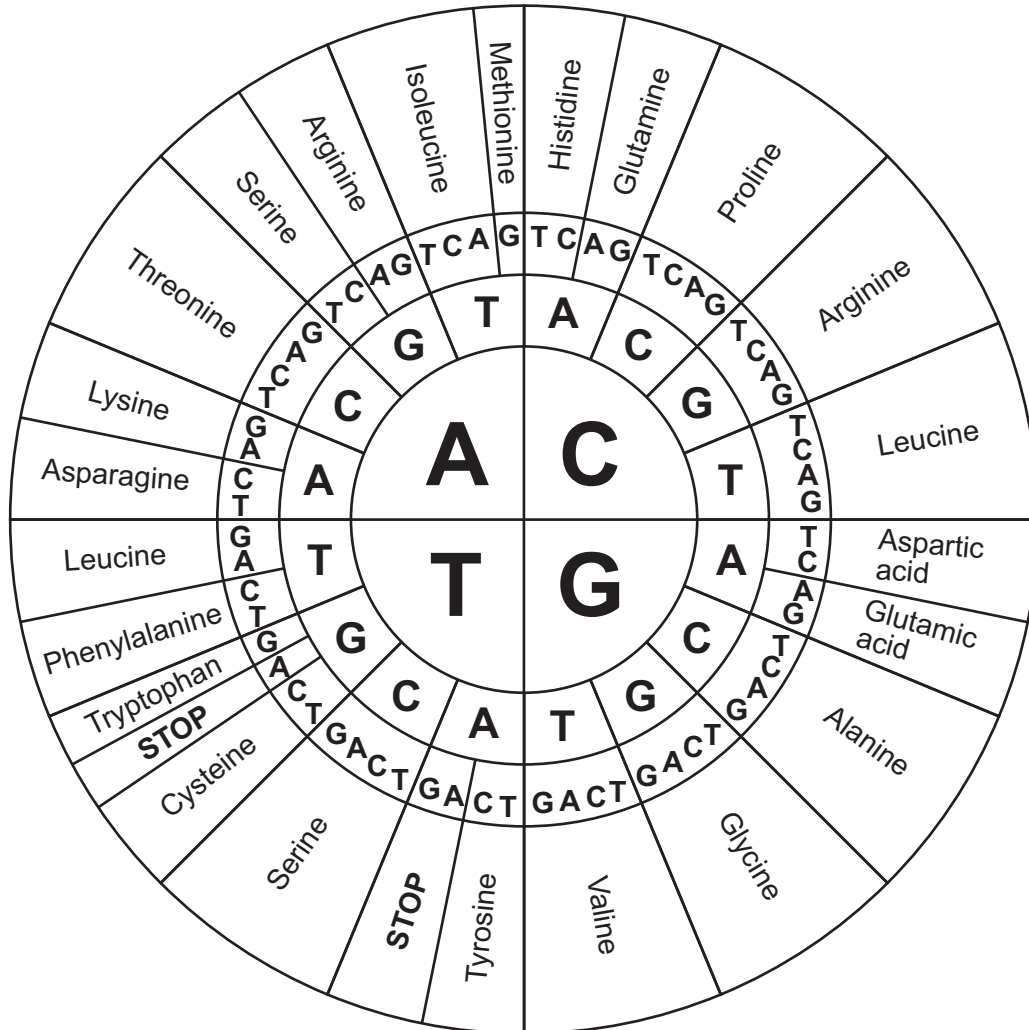


Fig. 5.3

Dipeptides are sometimes translated from TERRA RNA.

Use Fig. 5.3 to state the **two** amino acids in the dipeptide translated from TERRA RNA.

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- 2 [2]

- (iii) Scientists have discovered that TERRA interacts with genes in stem cells.

Increased concentrations of TERRA in stem cells result in a large increase in the number of genes that are transcribed.

Suggest the result of the large increase in the number of genes that are transcribed in stem cells.

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- [1]